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LID STRUCTURE FOR VEHICLE

Abstract

A lid structure for a vehicle includes a shaft member located on a rear side of an exterior surface that is visible from an outside; a lid that is swingable about the shaft member to open and close an opening formed on the exterior surface; and a support member that rotatably supports the shaft member. The support member includes a fixing portion fixed to the rear side of the exterior surface. When viewed in a direction along the shaft member, the fixing portion is located on a side opposite to the opening with the shaft member as a center.

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Background/Summary

FIELD OF THE INVENTION
[0001] The present invention relates to a lid structure for a vehicle.
BACKGROUND
[0002] A fuel filler port or a charging port of a vehicle may be opened and closed by a lid device referred to as a fuel lid. As a conventional technique related to such a lid device, there is a technique disclosed in European Patent No. 1508465. The lid device is configured such that an opening formed in a vehicle body is opened and closed by a swingable lid.
[0003] According to the lid device disclosed in the above-described specification, when the lid is in a fully open state, an end portion of the lid is located in the vicinity of a design surface. In this state, for example, a person may unintentionally come into contact with the lid and apply a force in a direction to further open the lid. Then, the end portion of the lid comes into contact with the design surface, thereby peeling the paint on the design surface or causing scratches to the design surface, which is a concern. Improvements are desired not only for fuel lids, but also from the perspective of maintaining a high level of aesthetic appearance.
[0004] An object of the present disclosure is to provide a lid structure for a vehicle capable of maintaining a high level of aesthetic appearance.
SUMMARY
[0005] According to the present disclosure, there is provided a lid structure for a vehicle including: a shaft member located on a rear side of an exterior surface that is visible from an outside; a lid that is swingable about the shaft member to open and close an opening formed on the exterior surface; and a support member that supports the shaft member. The support member includes a fixing portion fixed to the rear side of the exterior surface. When viewed in a direction along the shaft member, the fixing portion is located on a side opposite to the opening with the shaft member as a center.
[0006] According to the present disclosure, it is possible to provide the lid structure for a vehicle capable of maintaining a high level of aesthetic appearance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a front view of a lid device in which a lid structure for a vehicle according to an embodiment is adopted;
[0008] FIG. **2** is a cross-sectional view taken along line **2-2** in FIG. **1**;
[0009] FIG. **3** is a view when viewed in the direction of arrow **3** in FIG. **2**;
[0010] FIG. **4** is a view describing a state where which a lid illustrated in FIG. **2** is fully opened; and
[0011] FIG. **5** is an enlarged view of portion **5** in FIG. **4**.
DETAILED DESCRIPTION
[0012] An embodiment of the invention will be described below based on the accompanying drawings. A mode illustrated in the accompanying drawings is one example, and the invention is not limited to the mode.
EMBODIMENT
[0013] Referring to FIG. **1**, FIG. **1** illustrates a lid device **20** for a vehicle (hereinafter, referred to as the “lid device **20**”) that uses a lid structure for a vehicle. The lid device **20** is, for example, a fuel lid that opens and closes a fuel filler port or a charging port. The lid device **20** opens and closes an opening **11** formed in an exterior member **10** provided on a vehicle body.
[0014] The exterior member **10** is a component made of a resin material. The exterior member **10** is, for example, a front grille provided at the front of a vehicle. A surface on a vehicle exterior side of the exterior member **10** is referred to as an “exterior surface **12**” that is visible from the outside.

The opening **11** that is a substantially rectangular hole is formed in the exterior member **10**.

[0015] Referring to FIG. **2** and FIG. **3**, a plurality of bosses **14** and **15** for fixing the lid device **20** are formed on an exterior rear surface **13** that is a surface located on a rear side of the exterior surface **12**. The lid device **20** is fixed to the bosses **14** and **15** by screw fastening members **17** and **18** configured as screws.

[0016] Referring to FIG. **2**, the lid device **20** includes a support member **30** fixed to the rear side of the exterior member **10**; a shaft member **22** serving as a shaft supported by the support member **30**; and a lid **40** supported by the shaft member **22** and capable of opening and closing the opening **11**.

[0017] Referring to also FIG. **4**, the lid **40** is swingable from a fully closed position illustrated in FIG. **2** to a fully open position illustrated in FIG. **4**.

[0018] Referring to FIG. **1**, edges of the opening **11** having a substantially rectangular shape, which correspond to the respective sides, are referred to as a first edge portion **11***a*, a second edge portion **11***b*, a third edge portion **11***c*, and a fourth edge portion **11***d* in order from the edge on the left side of the figure.

[0019] Referring to also FIG. **2**, the first edge portion **11***a* and the third edge portion **11***c* extend substantially parallel to an extending direction of the shaft member **22**, and the second edge portion **11***b* and the fourth edge portion **11***d* extend substantially perpendicular to the extending direction of the shaft member **22**. The first edge portion **11***a* is located closer to the shaft member **22** than the third edge portion **11***c*, and the first edge portion **11***a* can be rephrased as a proximal edge **11***a* that is closest to the shaft member **22** among the first edge portion **11***a* to the fourth edge portion **11***d*. The third edge portion **11***c* is located farther from the shaft member **22** than the first edge portion **11***a*, and the third edge portion **11***c* can be rephrased as a distal edge **11***c*.

[0020] The shape of the opening **11** formed on the exterior surface **12** is not limited to a rectangular shape, and may be, for example, a circular shape. In the case of a circular opening, it is preferable that the "proximal edge" includes at least a portion of a circular edge which is closest to the shaft member **22**.

[0021] The second edge portion **11***b* can be rephrased as a first perpendicular edge **11***b* extending perpendicular to the first edge portion **11***a* and the third edge portion **11***c*. The fourth edge portion **11***d* can be rephrased as a second perpendicular edge **11***d* extending perpendicular to the first edge portion **11***a* and the third edge portion **11***c*.

[0022] Referring to FIG. **2**, of the bosses **14** and **15**, the boss **14** formed in a portion closer to the first edge portion **11***a* is referred to as a first boss **14**, and the boss **15** formed in a portion closer to the third edge portion **11***c* is referred to as a second boss **15**. The first boss **14** is closest to the shaft member **22**, and is farther from the proximal edge **11***a* than the shaft member **22**. The second boss **15** is formed at a position farther from the shaft member **22** than the first boss **14**.

[0023] Any type of screw such as a machine screw, a tapping screw, a cap bolt, or a bolt can be used as the screw fastening members **17** and **18**. Of the screw fastening members **17** and **18**, the screw fastening member **17** fastened to the first boss **14** is referred to as a first fastening member **17**, and the screw fastening member **18** fastened to the second boss **15** is referred to as a second fastening member **18**.

[0024] The support member **30** is configured as an integrally molded article. The support member **30** is a substantially box-shaped member that is open at a portion to which the lid **40** is attached. The support member **30** includes a first fixing portion **31** (fixing portion **31**) fixed to the first boss **14**; a second fixing portion **32** fixed to the second boss **15**; a first side wall portion **33** connected to an end portion of the first fixing portion **31** and extending in a direction away from the exterior member **10**; a second side wall **34** connected to an end portion of the second fixing portion **32** and extending in a direction away from the exterior member **10**; a support member contact portion **35** that extends from an end portion (one end) of the first side wall **33**, which is closer to the exterior member **10**, along the rear surface of the exterior member **10**, and that abuts against the exterior rear surface **13** of the exterior member **10**; a rear wall **36** that connects an end portion (the other end) of the first side wall **33**, which is farther from the exterior member **10**, to an end portion (the other end) of the second side wall **34**, which is farther from the exterior member **10**; a bottom portion **37** that closes a lower end; and a ceiling portion **38** (refer to FIG. **3**) that faces the bottom portion **37** and that closes an upper end.

[0025] Referring to FIG. **5**, the support member contact portion **35** includes a support member contact portion body **35***a* facing the exterior rear surface **13** and formed in an arch shape, and ribs **35***b* (support member-side ribs **35***b*) which extend from the support member contact portion body **35***a* toward the exterior rear surface **13**, and of which the tips can come into contact with the exterior rear surface **13**. A plurality of the ribs **35***b* are formed. Since the ribs **35***b* extend in the direction of an axis CL, deformation of the support member **30** in the direction of the axis CL can be suppressed.

[0026] Referring to FIG. **2**, the lid **40** includes a lid body **50** made of resin and supported by the shaft member **22**; a lid design portion **42** made of resin, supported by the lid body **50**, and exposed to the outside of the vehicle; and a held portion **43** extending from a tip of the lid body **50** toward the support member **30** and held by a lock device (not illustrated) in a fully closed state.

[0027] The lid body **50** includes a lid contact portion **51** that extends from the shaft member **22** in a radial direction and that can come into contact with the support member **30** when the lid **40** is in the fully open position, and a connecting portion **52** that connects a tip of the lid contact portion **51** to an end portion on a proximal edge **11***a* side of the lid design portion **42**.

[0028] Referring to FIG. **5**, the lid contact portion **51** includes a plurality of ribs **51***a* (lid-side ribs **51***a*) of which the tips can abut against the support member **30** in a fully open state.

[0029] When viewed in the direction along the axis CL of the shaft member **22**, the proximal edge **11***a*, the lid contact portion **51**, the shaft member **22**, and the fixing portion **31** are arranged in order with reference to a direction from the proximal edge **11***a* toward the first fixing portion **31**, and the support member contact portion **35** and the lid contact portion **51** are spaced apart from the proximal edge **11***a*. In other words, the lid contact portion **51** (ribs **51***a*) and the shaft member **22** are located closer to a fixing portion **31** side than the proximal edge **11***a*. The lid contact portion **51** is located between the proximal edge **11***a* and the shaft member **22**. In addition, the lid contact portion **51** and the support member contact portion **35** overlap each other in a direction orthogonal to the exterior surface **12**.

[0030] The connecting portion **52** is formed in a curved shape so as not to abut against the proximal edge **11***a* when the lid **40** is displaced from the fully closed state to the fully open state. Furthermore, the lid contact portion **51** (ribs **51***a*) also does not abut against the proximal edge **11***a* when the lid **40** is displaced from the fully closed state to the fully open state. When the lid **40** that is in the fully open position is swung in a direction to further open the lid **40**, the direction of a force applied to the support member **30** is substantially orthogonal to the exterior surface **12**, a force in the opposite direction acts on the fixing portion **31** on a side opposite to the shaft member **22**, so that the lid **40**, the support member **30**, and the exterior surface **12** move to rotate about the shaft member **22**. Therefore, the lid **40** and the exterior surface **12** are less likely to come into contact with each other, so that the exterior surface **12** can be prevented from being scratched.

[0031] Referring to FIG. **1** and FIG. **2**, a lid design surface **42***a* of the lid design portion **42**, which is a surface exposed to the outside of the vehicle, is substantially continuous with the exterior surface **12** when the lid **40** is in the fully closed position. The lid design surface **42***a* has a substantially rectangular shape along the opening **11**.

[0032] A part of the lid design surface **42***a* may be formed as a recess **42***b* formed in a slightly recessed shape compared to other portions. Accordingly, an operator can easily recognize a portion that has to be operated.

[0033] Referring to FIG. **5**, when the lid **40** is in the fully open state, at least the end portion on the proximal edge **11***a* side of the lid design portion **42** is located at a portion farther from the proximal edge **11***a* than the shaft member **22**.

[0034] The edges of the opening **11** are formed integrally with an opening protrusion **11***e* protruding toward the inside of the vehicle.

[0035] The lid structure for a vehicle adopted in the lid device **20** described above will be summarized below.

[0036] Referring to FIG. **2**, in a first aspect, the lid structure for a vehicle includes the shaft member **22** located on the rear side of the exterior surface **12** that is visible from the outside; the lid **40** that is swingable about the shaft member **22** to open and close the opening **11** formed on the exterior surface **12**; and the support member **30** that supports the shaft member **22**. The support member **30** includes the first fixing portion **31** (fixing portion **31**) fixed to the rear side of the exterior surface **12**. When viewed in the direction along the shaft member **22** (with reference to a cross section extending in a circumferential direction with the shaft member **22** as a center), the first fixing portion **31** is located on the side opposite to the opening **11** with the shaft member **22** as a center.

[0037] Referring to FIG. **5**, when the lid **40** that is in the fully open position is swung in the direction to further open the lid **40** as indicated by arrow **61**, a portion of the first fixing portion **31** of the support member **30**, the portion including the exterior surface **12**, is pulled toward the rear side of the exterior surface **12** as indicated by arrow **62**. Since the exterior surface **12** is displaced in a movement direction of the lid **40** in accordance with the displacement of the support member **30**, the lid **40** and the exterior surface **12** are less likely to come into contact with each other, and the exterior surface **12** can be prevented from being scratched. It is possible to provide the lid structure for a vehicle capable of maintaining a high level of aesthetic appearance.

[0038] In a second aspect, in the lid structure for a vehicle according to the first aspect, the lid **40** includes the lid contact portion **51** capable of coming into contact with the support member **30** when the lid **40** is in the fully open position where the lid **40** is fully open. By bringing the lid contact portion **51** into contact with the support member **30**, the direct application of a load to the portion including the exterior surface **12** is avoided.

[0039] In a third aspect, in the lid structure for a vehicle according to the second aspect, the lid contact portion **51** includes a rib **51***a* of which the tip is capable of coming into contact with the support member **30**. The strength of the lid **40** can be increased by the rib **51***a*.

[0040] In a fourth aspect, in the lid structure for a vehicle according to any one of the first to third aspects, the support member **30** includes the support member contact portion **35** in contact with the rear side of the exterior surface **12**. When a portion of the edge of the opening **11** of the exterior surface **12**, the portion being closest to the shaft member **22**, is defined as the proximal edge **11***a*, the support member contact portion **35** includes a rib **35***b*, of which the tip is capable of coming into contact with the rear side of the exterior surface **12**, between the shaft member **22** and the proximal edge **11***a*. The strength of the support member **30** around the shaft member **22** is increased.

[0041] In a fifth aspect, in the lid structure for a vehicle according to the fourth aspect, the lid **40** includes the lid contact portion **51** capable of coming into contact with the support member **30** when the lid **40** is in the fully open position where the lid **40** is fully open. The support member contact portion **35** is sandwiched between the lid contact portion **51** and the rear side of the exterior surface **12** on the rear side of the exterior surface **12** when the lid **40** is in the fully open position. By causing the lid contact portion **51** to abut against a portion with high strength, the life span of the lid device **20** to which the lid structure for a vehicle is applied can be extended.

[0042] In a sixth aspect, in the lid structure for a vehicle according to any one of the first to fifth aspects, when viewed in the direction along the axis CL of the shaft member **22**, the proximal edge **11***a*, the lid contact portion **51**, the shaft member **22**, and the first fixing portion **31** are arranged in order with reference to the direction from the proximal edge **11***a* toward the fixing portion **31**. The support member contact portion **35** and the lid contact portion **51** are spaced apart from the proximal edge **11***a*. In other words, the lid contact portion **51** (rib **51***a*) and the shaft member **22** are located closer to the fixing portion **31** side than the proximal edge **11***a*. The lid contact portion **51** is located between the proximal edge **11***a* and the shaft member **22**. A load on the lid **40** acts as a force that pulls the exterior surface **12** in the vicinity of the fixing portion **31** toward the rear side of the exterior surface **12** with the shaft member **22** as an axis.

[0043] In a seventh aspect, in the lid structure for a vehicle according to any one of the first to sixth aspects, the exterior surface **12** is a part of the exterior member **10** (for example, a front grille) that is fixable to the vehicle body. Since the exterior member **10** can be fixed to the vehicle body, the support member **30** can be fixed to the exterior member **10** in advance, and then the exterior member **10** can be fixed to the vehicle body.

[0044] The exterior member **10** (exterior surface **12**) is not recessed from an opening end into the opening (in the direction away from the exterior surface **12**). In other words, the exterior member **10** does not wrap around to a rear side of the lid **40** (lid design portion **42**), and does not overlap the lid **40**. Therefore, the shaft member **22** can be brought closer to the exterior surface **12**. Accordingly, the connecting portion **52** of the lid **40** does not become large and can be connected near the end of the lid design portion **42**, so that high rigidity of the lid **40** can be ensured. In addition, when the connecting portion **52** is made compact, the amount of bending of the lid **40** can be suppressed, and interference with and scratches to the exterior surface **12** can also be prevented.

[0045] In an eighth aspect, in the lid structure for a vehicle according to any one of the first to seventh aspects, the support member **30** includes the first fixing portion **31** fixed to the exterior member **10**. The first fixing portion **31** is fixed to the exterior member **10** by the first screw fastening member **17** (screw fastening member **17**).

[0046] Screw fastening provides high axial force and excellent pulling force for the exterior surface **12** (when the support member **30** is shifted in the direction of the rear side of the exterior surface **12**, the exterior surface **12** bends in the direction of the rear side, together with the support member **30**). Therefore, interference between the exterior surface **12** and the lid **40** can be prevented, and the exterior surface **12** can be prevented from being scratched. In addition, screw fastening can be performed more easily than other fastening methods such as welding.

[0047] An example in which the lid structure for a vehicle according to the invention is applied to a fuel lid has been described; however, the lid device **20** is not limited to being provided in the exterior member **10** fixed to the vehicle body, and may be provided in a component inside a passenger compartment. When the lid device **20** is provided inside the passenger compartment, the outside from which the exterior surface **12** is visible refers to a surface that is visible to occupants inside the passenger compartment.

[0048] The invention is not limited to the embodiment as long as the actions and effects of the invention are exhibited.

Claims

1. A lid structure for a vehicle, comprising: a shaft member located on a rear side of an exterior surface that is visible from an outside; a lid that is swingable about the shaft member to open and close an opening formed on the exterior surface; and a support member that supports the shaft member, wherein the support member includes a fixing portion fixed to the rear side of the exterior surface, and when viewed in a direction along the shaft member, the fixing portion is located on a side opposite to the opening with the shaft member as a center.

2. The lid structure for a vehicle according to claim 1, wherein the lid includes a lid contact portion capable of coming into contact with the support member when the lid is in a fully open position where the lid is fully open.

3. The lid structure for a vehicle according to claim 2, wherein the lid contact portion includes a rib of which a tip is capable of coming into contact with the support member.

4. The lid structure for a vehicle according to claim 1, wherein the support member includes a support member contact portion in contact with the rear side of the exterior surface, and when a portion of an edge of the opening of the exterior surface, the portion being closest to the shaft member, is defined as a proximal edge, the support member contact portion includes a rib, of which a tip is capable of coming into contact with the rear side of the exterior surface, between the shaft member and the proximal edge.

5. The lid structure for a vehicle according to claim 2, wherein the support member includes a support member contact portion in contact with the rear side of the exterior surface, and when a portion of an edge of the opening of the exterior surface, the portion being closest to the shaft member, is defined as a proximal edge, the support member contact portion includes a rib, of which a tip is capable of coming into contact with the rear side of the exterior surface, between the shaft member and the proximal edge.

6. The lid structure for a vehicle according to claim 3, wherein the support member includes a support member contact portion in contact with the rear side of the exterior surface, and when a portion of an edge of the opening of the exterior surface, the portion being closest to the shaft member, is defined as a proximal edge, the support member contact portion includes a rib, of which a tip is capable of coming into contact with the rear side of the exterior surface, between the shaft member and the proximal edge.

7. The lid structure for a vehicle according to claim 5, wherein the support member contact portion is sandwiched between the lid contact portion and the rear side of the exterior surface when the lid is in a fully open position.

8. The lid structure for a vehicle according to claim 6, wherein the support member contact portion is sandwiched between the lid contact portion and the rear side of the exterior surface when the lid is in a fully open position.

9. The lid structure for a vehicle according to claim 5, wherein when viewed in the direction along the shaft member, the proximal edge, the lid contact portion, the shaft member, and the fixing portion are arranged in order with reference to a direction from the proximal edge toward the fixing portion, and the support member contact portion and the lid contact portion are spaced apart from the proximal edge.

10. The lid structure for a vehicle according to claim 6, wherein when viewed in the direction along the shaft member, the proximal edge, the lid contact portion, the shaft member, and the fixing portion are arranged in order with reference to a direction from the proximal edge toward the fixing portion, and the support member contact portion and the lid contact portion are spaced apart from the proximal edge.

11. The lid structure for a vehicle according to claim 7, wherein when viewed in the direction along the shaft member, the proximal edge, the lid contact portion, the shaft member, and the fixing portion are arranged in order with reference to a direction from the proximal edge toward the fixing portion, and the support member contact portion and the lid contact portion are spaced apart from the proximal edge.

12. The lid structure for a vehicle according to claim 8, wherein when viewed in the direction along the shaft member, the proximal edge, the lid contact portion, the shaft member, and the fixing portion are arranged in order with reference to a direction from the proximal edge toward the fixing portion, and the support member contact portion and the lid contact portion are spaced apart from the proximal edge.

13. The lid structure for a vehicle according to claim 1, wherein the exterior surface is a part of an exterior member that is fixable to a vehicle body.

14. The lid structure for a vehicle according to claim 13, wherein the fixing portion is fixed to the exterior member by a screw fastening member.